MILITARY SPECIFICATION SHEET

ELECTRON TUBE, NEGATIVE GRID (MICROWAVE)

TYPE 7077

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Triode, planar, metal-ceramic

See figure 1

Mounting position: Any

Weight: 0.06 ounce (1.7 grams) nominal

ABSOLUTE RATINGS:

Parameter:	F	Ef	Eb	Ebb	Ec	Ehk	Rk	Rg	Rp	Ik	Pp	TE	Barometric pressure, reduced
Unit:	MHz	V	Vdc	Vdc	Vdc	v	Ohms	Meg	Meg	mAdc	W	°C	mmHg
Maximum:	450	6.6	250	---	0,-50	50	---	0.01 (Note 1)	---	11.0	1.1	250 (Note 2)	8
Minimum:	---	6.0	---	---	---	---	---	---	---	---	---	---	---
TEST CONDITIONS:	---	6.3	---	250	0	---	82	(J) ---	0.018	---	---	---	---

GENERAL:

Qualification - Required

(J) denotes changes

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
(J)	Qualification inspection								
(J)1002	Barometric pressure, reduced	-	Pressure = 8 ±1 mmHg; 300 Vac applied between grid and anode	---	---	---	---	---	---
(J)1031	Sweep-frequency vibration	11	Ebb = 150 Vdc; Rp = 10,000 ohms; 10 G; F = 100 to 5,000 Hz	---	---	Ep	---	15	mVac
(J)1031	Sweep-frequency vibration fatigue	12	Eb = Ec = 0; Ef = 6.3 V	---	---	---	---	---	---
(J)---	Sweep-frequency vibration-fatigue test end points:								
1031	Low-frequency vibration	-		---	---	Ep	---	15	mVac
1336	Heater-cathode leakage	-		---	---	Ihk	---	20	µAdc
---	Noise figure (1)	-		---	---	NF	---	8.1	dB
---	Power gain	-		---	---	Gain	11.5	---	dB
	Quality conformance inspection, part 1								
1301	Heater current	-		0.65	II	If	222	258	mA
1336	Heater-cathode leakage	-		0.65	II	Ihk	---	20	µAdc
---	Noise figure (1)	3	F = 450 MHz; Cp = 500 pF	0.65	II	NF	---	6.6	dB
---	Power gain	4	F = 450 MHz; Cp = 500 pF	0.65	II	Gain	12.5	---	dB
1201	Short and discontinuity detection	-		0.65	II	---	---	---	---
	Quality conformance inspection, part 2								
1211	Insulation of electrodes	-		---	---	R	100	---	Meg
1256	Electrode current (anode)	-		---	---	Ib	5.2	8.8	mAdc
1266	Grid emission	6	Ef = 7.0 V; Ecc = -20 Vdc; Rg = 0.1 Meg;	---	---	Ic	0	-2.0	µAdc
2210	Grid recovery	5	Ebb = 250 Vdc; Ec/Ib = 3.0 mAdc; Rp = 10 K ohms	---	---	ΔIb Δib	--- ---	0.6 1.0	mAdc ma
1306	Transconductance (1)	-	Cp = 4 µF; Ck = 1,000 µF	---	---	Sm	7,000	13,000	µmhos
1306	Transconductance (2)	-	Ef = 6.0 V; Cp = 4 µF; Ck = 1,000 µF	---	---	ΔSm Ef	---	20	%
---	Noise figure (2)	3	Ef = 6.0 V; F = 450 MHz; Cp = 500 pF	---	---	NF	---	8.1	dB
1316	Amplification factor	-		---	---	Mu	65	115	---
1331	Direct-interelectrode capacitance	-	DESC Drawing No. 67030	---	---	Cgp Cin Cout Chk	1.06 1.20 0.005 0.80	1.36 2.10 0.015 1.40	pF pF pF pF
---	Envelope strain	7		---	---	---	---	---	---

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS		UNIT
							MIN	MAX	
	Quality conformance inspection, part 2 -Continued								
(J)1031	Low-frequency vibration	10	Ebb = 150 Vdc; Rp = 10,000 ohms; 15 G; F = 40 Hz	---	---	Ep	---	10	mVac
	Quality conformance inspection, part 3								
1506	Heater-cycling life	8	Ef = 7.0 V; Ehk = 70 Vdc; Rk = 0; Ec = Eb = 0	---	---	---	---	---	---
---	Heater-cycling life-test end point:								
1336	Heater-cathode leakage	-		---	---	Ihk	---	40	µAdc
1516	Stability life	-	Ebb = 300 Vdc; Ehk = 70 Vdc; Rb = 0.018 Meg; Rk = 82 ohms; Rg = 0.01 Meg	2.5	Code F	---	---	---	---
---	Stability life-test end point:								
---	Change in trans-conductance (1) of individual tubes	-		---	---	ΔSm t	---	15	%
1501	Intermittent life (1)	9	Group A; stability-life test, or equivalent conditions; t = 1,000 hours	---	---	---	---	---	---
(J)1501	Intermittent life (2)	9-13	Stability-life test, or equivalent conditions, except TE = 250°C (min); t = 1,000 hours	---	---	---	---	---	---
---	Intermittent life-test end points:								
1301	Heater current	-		---	---	If	222	264	mA
---	Noise figure (1)	-		---	---	NF	---	8.6	dB
---	Noise figure (2)	-		---	---	NF	---	10.5	dB
---	Power gain	-		---	---	Gain	11.0	---	dB
1336	Heater-cathode leakage	-		---	---	Ihk	---	20	µAdc
1211	Insulation of electrodes	-		---	---	R	50	---	Meg
1306	Transconductance (1)	-		---	---	Sm	6,000	---	µmhos
1041	Shock	13	Hammer angle = 30°; Ehk = 50 Vdc; Eb = 150 Vdc; Rp = 0	---	---	---	---	---	---
	Shock-test end points:								
1031	Low-frequency vibration	-		---	---	Ep	---	15	mVac
1336	Heater-cathode leakage	-		---	---	Ihk	---	20	µAdc
---	Noise figure (1)	-		---	---	NF	---	8.1	dB
---	Power gain	-		---	---	Gain	11.5	---	dB

NOTES:

1. At anode temperature below 150°C, Rg may be increased to 0.1 Meg.
2. Envelope temperature is defined as the temperature measured at the periphery of the anode.

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NOTES: -Continued

3. Noise figure test.

(a) Definition. The noise figure of the system shall be defined as 10 log $\frac{32.4}{Y-1}$, where 32.4 is the excess noise temperature of the noise generator, expressed as power ratio, and Y is the ratio of the noise power output of the system with the noise generator "on" to the noise power output of the system with the noise generator "off". The specified test limits refer to the noise figure of the tube under test (TUT) which is the noise figure of the system corrected for the noise figure of the noise amplifier and of the first stage.

The first stage noise figure is calculated with the formula:

$F_1 = F_{1-2} - \frac{(F_2 - 1)}{G_1}$ where F_1 is the noise figure of the first stage, F_2 is the noise figure of the noise figure amplifier, F_{1-2} is the measured noise figure of the overall systems (all noise figures being expressed as power ratios rather than in decibels) and G_1 is the average gain of the TUT expressed as a power ratio.

(b) Test procedure. Measurement of noise figure is made by noting the noise power in the calibrated power indicator both with the noise generator switched "on" and with it switched "off", using the equipment shown on figure 2.

(c) Noise generator. An Airborne Instrument Laboratory Argon Lamp Noise Generator, Model 70, or equivalent.

(d) Four hundred and fifty MHz test amplifier. An amplifier consisting of the TUT with related circuits shown on figure 3.

(1) The cathode cavity shall be of sufficient quality to maintain an unloaded Q of over 150 measured at the amplifier input with the tube removed and a low-loss dummy-tube capacitor having a value within the range specified for input capacitance inserted in its place. The heater circuit shall also be a component part of the cathode cavity such that the unloaded Q will reflect any loss in input power due to the heater circuit. The cathode cavity and its components shall provide a power match between the coaxial input and the cathode of the TUT.

(2) The anode cavity shall be of sufficient quality to maintain an unloaded Q in excess of 600 measured with the TUT replaced with a low-loss dummy-tube capacitor having a value within the range specified herein for output capacitance. The anode cavity shall be capable of resonating all tubes having output capacitance within the limits specified herein. An appropriate coupling loop or probe shall be provided to load the anode cavity in order to maintain an overall amplifier bandwidth of 7.5 MHz when connected to its 50-ohm load.

(3) The design of the tube contacting components shall provide for firm and consistent contacts to all tube terminals. Sufficient shielding between input and output circuits of the amplifier shall be provided so that no indication is obtained on the power gain test indicator when the dummy-type capacitor is substituted for the TUT and full signal is applied. Both input and output resonant circuits shall be tuned to 450 MHz for shielding test.

(e) Four hundred and fifty MHz noise amplifier. The amplifier shall have the following characteristics:

Center frequency	- 450 MHz
Bandwidth	- 2.0 MHz
Input VSWR from 50 ohms	- Less than 1.1
Noise figure	- Less than 6 dB (using Airborne Instrument Laboratory Argon Lamp Noise Generator, Model 70, or equivalent).
Power gain	- Approximately 80 dB (shall be adjustable from 70 to 80 dB). The output shall be adjusted to work into a calibrated 50-ohm load and shall be matched to a VSWR less than 1.2 looking into noise amplifier output.

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NOTES: -Continued

(f) Detector and 50-ohm load. The detector and load shall be the equivalent of the General Radio 874 VQ detector terminated in a 50-ohm load (General Radio 874 WM). A calibration curve shall be plotted to accurately establish the relation between rf power into the detector and the detector crystal current as indicated on the microampere meter. The calibration curve is used to accurately determine the Y factor in the noise figure equation.

(g) Microampere meter. A Greibach, Model 500, Microampere Meter, or equivalent.

(h) Dummy-type capacitor. A dymmy-type capacitor is a capacitor used in checking the test equipment and in making the test for adequate shielding between input and output circuits of the 450 MHz test amplifier TUT. Physical dimensions of the capacitor shall be the same as for the tube except that the heater pins and the heater-to-cathode ceramic are omitted. Construction shall be as shown on figure 4.

4. Power gain test.

(a) The power gain test shall be performed in the equipment as shown on figure 5.

(1) Calibrated signal generator - General Radio Model 1021-P2, or equivalent.

(2) Four hundred and fifty MHz noise amplifier - Same as specified for noise figure tests. (See note 3(e).)

(3) Detector and 50-ohm load - Same as specified for noise figure test. (See note 3(f).)

(4) Microampere meter - Same as specified for noise figure test. (See note 3(g).)

(b) Procedure. With signal generator switched directly into detector and 50-ohm load, adjust output for approximate midscale indication on microampere meter and note output, P1 (calibrated in decibels), from signal generator. Switch 450-MHz test amplifier into circuit with tube inserted and reduce signal generator output until the same approximate midscale indication is again obtained on the microampere meter and note the signal generator output, P2. Power gain = P1 -P2.

5. Grid recovery. Test shall be performed as follows:

(a) With TUT operating under specified conditions, adjust Ec for Ib = 3 mAdc.

(b) Apply 5 volt pulse, 20 µs duration, 60 pps to the grid.

(c) With application of pulse, measure undershoot (Δib) and change in average current (ΔIb).

6. Prior to this test, tubes shall be preheated for a minimum of 5 minutes at the conditions indicated below. The 3-minute test is not permitted. Test within 3 seconds after preheating.

Ef	Ecc	Ebb	Rk	Rg	Rp
V	Vdc	Vdc	Ohms	Meg	Meg
7.0	0	250	82	0.1	0.018

7. Envelope strain procedure. Tubes shall be tested as specified in method 2126, except that they shall be immersed in water at not more than 5°C for 15 seconds and immediately thereafter subjected to the standard temperature cycle specified in method 2126. The criteria for air leaks shall be heater current of 300 mA, or more.

8. The heater-cycling life-test sample shall consist of 15 tubes and no tube failure shall be permitted. In the event of rejection of the first sample due to failure of 1 tube, a second sample of 15 tubes shall be selected from the lot. Acceptance shall then be based on the combined first and second samples. The total tube failures from the combined first and second samples shall not exceed one.

9. Intermittent life. The specified life-test end points shall be applied to both the intermittent life-test (1) and (2) test samples with the exception of noise figure and power gain measurements, which shall be performed on the intermittent life-test (1) samples only.

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NOTES: -Continued

10. Low-frequency vibration. Test shall be performed as follows: The tube shall be vibrated with simple harmonic motion in each of two planes; first, parallel to the cylindrical axis; second, perpendicular to the cylindrical axis and parallel to a line through the heater contacts.

11. Sweep-frequency vibration. Test shall be performed as follows:

    (a) The frequency shall be increased from 100 to 5,000 Hz with approximate logarithmic progression in 3 ± 1 minutes. The return sweep (5,000 to 100 Hz) is not required.

    (b) The tube shall be vibrated with simple harmonic motion in each of two planes; first, parallel to the cylindrical axis; second, perpendicular to the cylindrical axis and parallel to a line through the heater contacts.

    (c) The value of alternating voltage, Ep, produced across the resistor, Rp, as a result of vibration shall be measured with a suitable device having a response to the rms value of a sine wave of voltage to within ±0.5 dB of the response at 100 Hz over the frequency range of 50 to 5,000 Hz and having a frequency cutoff such that the response is down a minimum of 12 dB at 10,000 Hz. The meter shall have a dynamic response characteristic equivalent to, or faster than, a VU meter (operated in accordance with Standard ASA No. C16.5-1954).

12. Sweep-frequency vibration fatigue. Test shall be performed as specified in method 1031, except that the tubes shall be vibrated for a total of 6 hours, that is, 3 hours in each of two directions: first, parallel to the cylindrical axis; second, perpendicular to the cylindrical axis and parallel to a line through the heater contacts.

13. This test shall be performed during the initial production and once each succeeding 12-calendar months in which there is production. A regular double sampling plan shall be used, with the first sample of three tubes with an acceptance number of zero, and a second sample of three tubes with a combined acceptance number of one. In the event of failure, the test will be made as a part of quality conformance inspection, part 2, code level D, with an AQL of 6.5. The regular "12-calendar month" double sampling plan shall be reinstated after three consecutive samples have been accepted.

Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Army - MI
Air Force - 80
DSA - ES

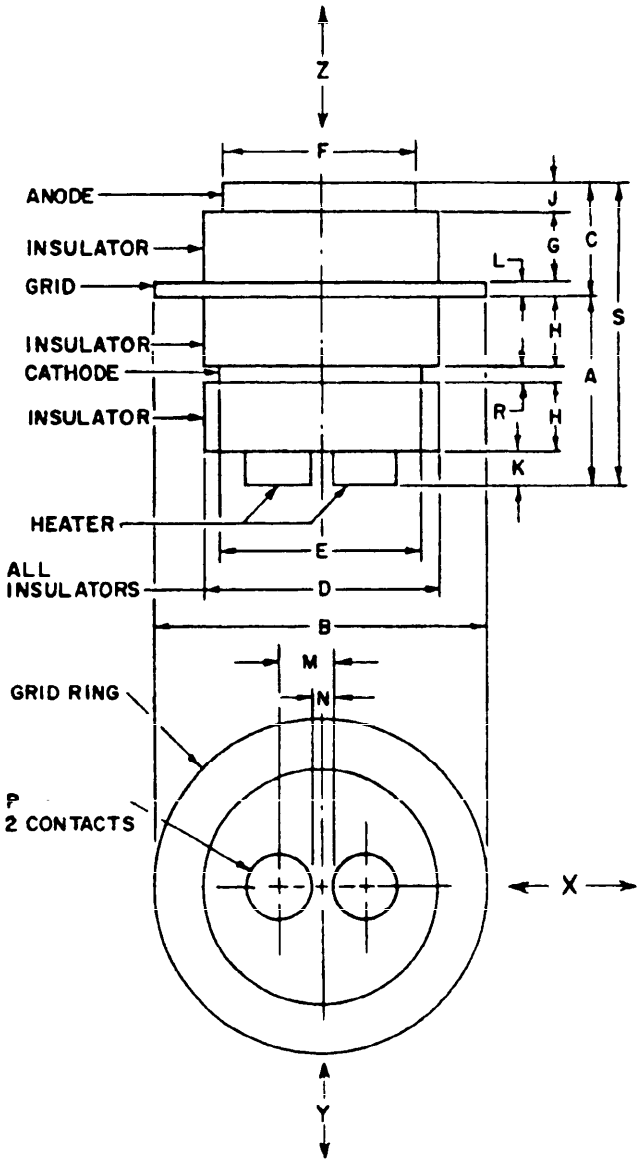
User activities:
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Navy - AS, OS, MC, CG
Air Force - 11

Preparing activity:
Navy - EC

Agent:
DSA - ES

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Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Quality conformance inspection, part I (see note e)		
A	.268 (6.81)	.292 (7.42)
B	.476 (12.09)	.484 (12.29)
C	.156 (3.96)	.174 (4.42)
D		.335 (8.51)
E	.281 (7.14)	.289 (7.34)
F	.271 (6.88)	.279 (7.09)
G	.094 (2.39)	.104 (2.64)
H	.095 (2.41)	.105 (2.67)
J	.034 (.86)	.046 (1.17)
K	.047 (1.19)	.063 (1.60)
L	.024 (.61)	.030 (.76)
M	.055 (1.40)	.081 (2.06)
N	.030 (.76)	
P	.086 (2.18)	.094 (2.39)
R	.022 (.56)	.028 (.71)
S	.430 (10.92)	.460 (11.68)

NOTES:
a. Eccentricity of anode, grid, and cathode with respect to centerline shall be .005 (.13 mm) maximum.
b. Eccentricity of insulators with respect to centerline shall be .010 (.25 mm) maximum.
c. Centerline of grid shall be reference line for horizontal tolerances.
d. Bottom surface of grid shall be reference plane for vertical tolerances.
e. The AQL for dimensions listed under quality conformance inspection, part I, shall be 1.0 percent, inspection level II.

FIGURE 1. Outline drawing of electron tube type 7077.

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FIGURE 2. Equipment set up for noise figure test.

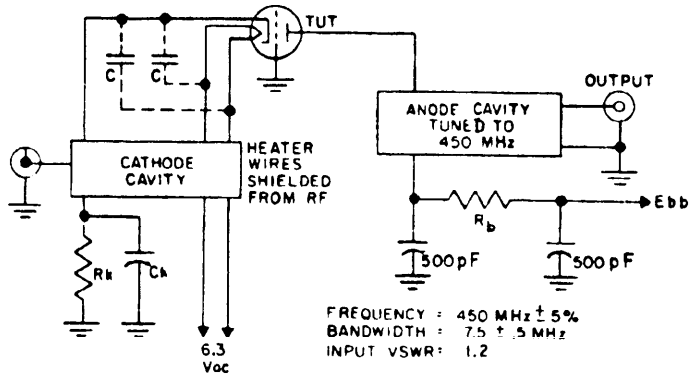


FIGURE 3. 450-MHz test amplifier.

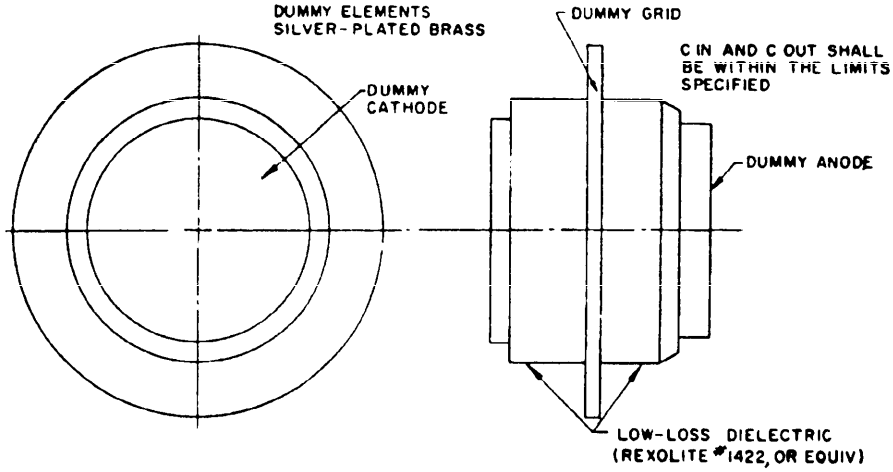


FIGURE 4. Construction of 'dummy-tube' capacitor.

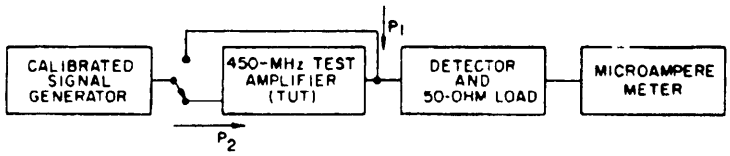


FIGURE 5. Equipment set up for power gain test.

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